Clarity from Chaos

AI-Powered Multi-Agent Platform for Unstructured Data Intelligence

Multi-Agent AI | RAG · Vector Search | Real-time SSE Streaming | LLM Insight Synthesis

Hackathon · AI & Data Track · TCS · 2025

PROBLEM STATEMENT

The Data Chaos Challenge

Organisations drown in siloed, multi-format data with no unified way to extract insight.

03
Cross-Document Conflicts

· Same metric reported differently
· across documents or time periods
· No automated conflict detection

04
No Actionable Intelligence

· Raw data ≠ business insight
· No natural-language Q&A layer
· No executive-ready summaries

01
Multi-Format Fragmentation

· PDFs, DOCX, CSV, JSON, images
· and video — all separate silos
· No single pipeline handles all

02
Manual & Slow Analysis

· Analysts spend 80 % of time
· cleaning and locating data
· Insights arrive too late

SOLUTION

One Platform. Any Format. Instant Insight.

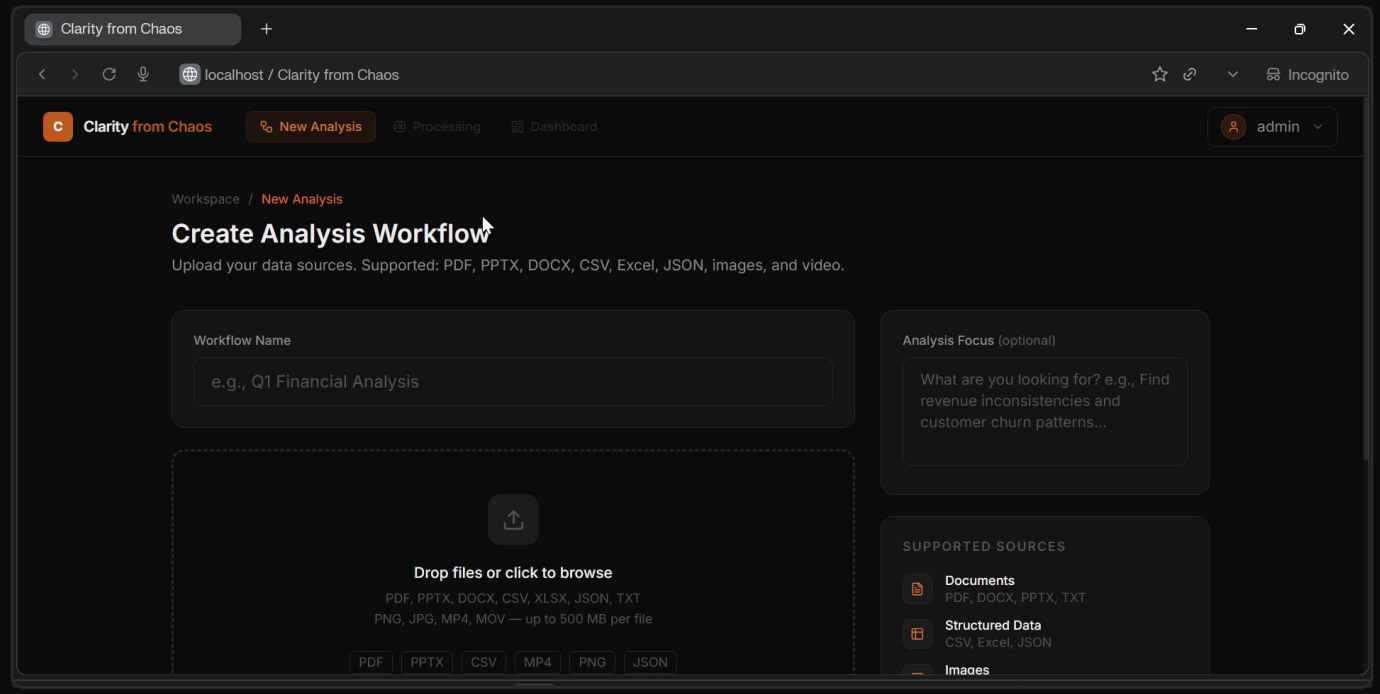
Universal Ingestion
Upload PDFs, DOCX, CSV, JSON, TXT in a single drag-and-drop workflow.

Multi-Agent Pipeline
10 specialised agents orchestrated end-to-end — parse, embed, extract, reason.

Cross-Doc Intelligence
Automatically detects conflicts and overlaps across multiple source documents.

LLM-Powered Insights
Synthesises facts into executive insights with evidence and confidence scores.

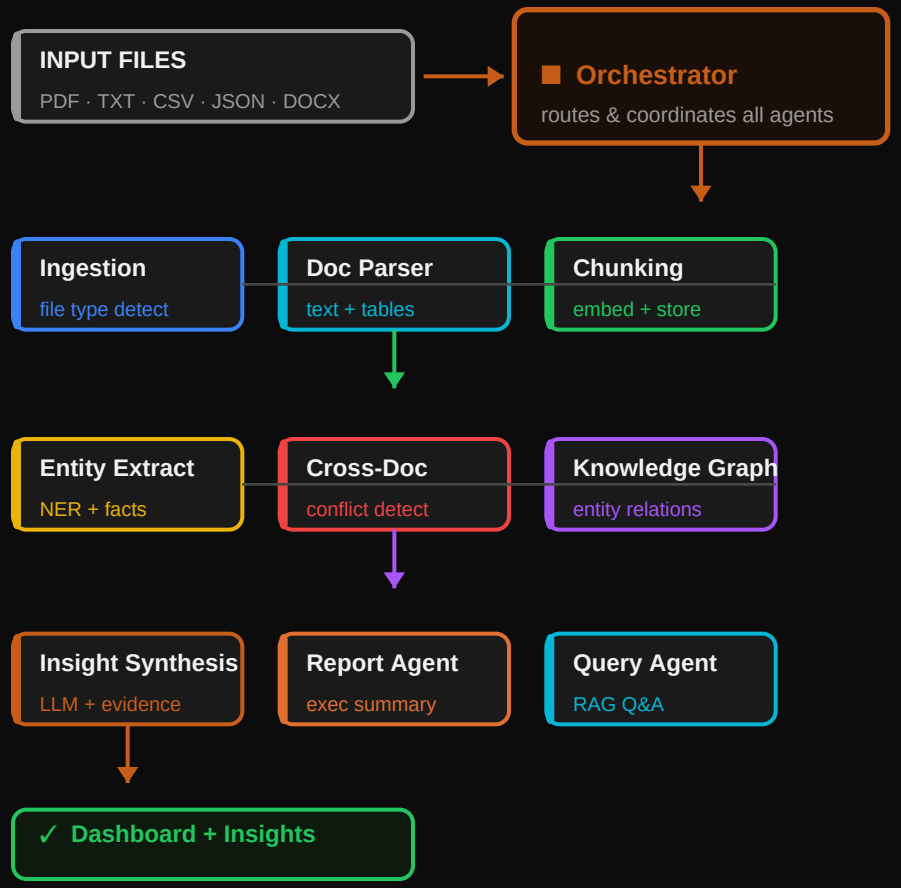
Natural Language Q&A
Ask plain-English questions over all ingested documents via RAG retrieval.



Create Analysis Workflow

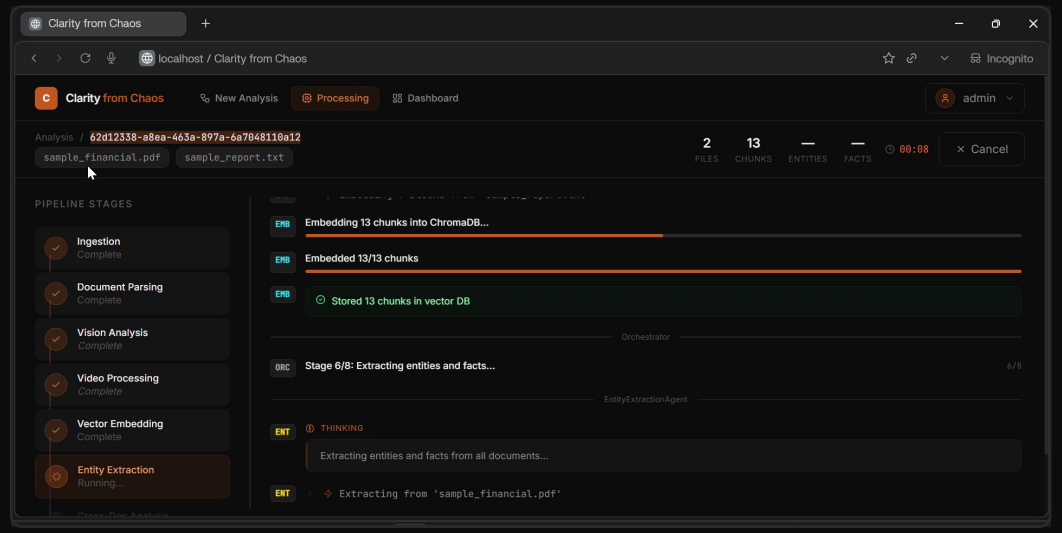
ARCHITECTURE

Multi-Agent Orchestration Pipeline



Key Innovations

- Cross-document conflict detection via fuzzy matching + LLM adjudication
- Knowledge graph built from fact-entity relationships (NetworkX)
- Full SSE event bus with history replay for live agent streaming
- RAG retrieval using sklearn HashEmbedder — zero native deps
- Orchestrator skips irrelevant stages based on uploaded file types



Live pipeline execution

TECH STACK

Built Entirely on Open-Source & Free Tier

Zero paid APIs required for core functionality — production-ready stack

BACKEND

Python 3.13 + Flask
REST API + SSE streaming

LangChain 0.3
LLM orchestration & chaining

OpenRouter
Free LLM gateway (GPT-4o-mini)

VECTOR & NLP

ChromaDB 0.6
Local persistent vector store

sklearn HashVectorizer
Pure-Python embeddings

spaCy + RapidFuzz
NER + fuzzy entity merging

DATA PROCESSING

pdfplumber + pypdf
PDF text & table extraction

pandas + NetworkX
Tabular data + graph engine

python-docx / pptx
Office document parsing

FRONTEND

React 18 + Vite
SPA with fast HMR dev server

Tailwind CSS
Utility-first dark theme UI

EventSource API
Real-time SSE agent streaming

STORAGE

SQLite
Jobs, insights, entities, facts

ChromaDB on disk
Vector embeddings per job

Local filesystem
Uploaded files — no cloud needed

DEPLOYMENT

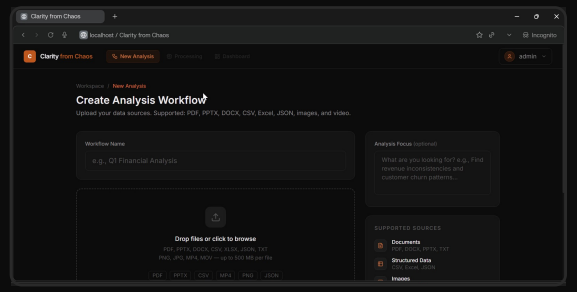
Vite proxy → Flask :5000
Single-origin dev setup

Python venv
Isolated dependency management

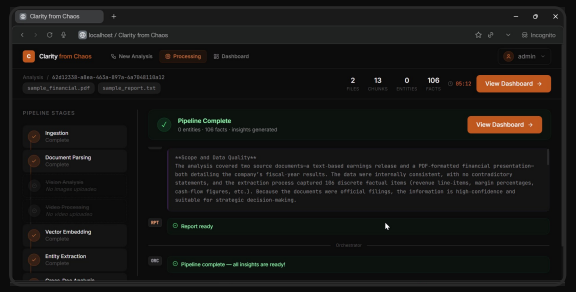
No Docker needed
Runs on plain Windows / Mac / Linux

LIVE DEMO

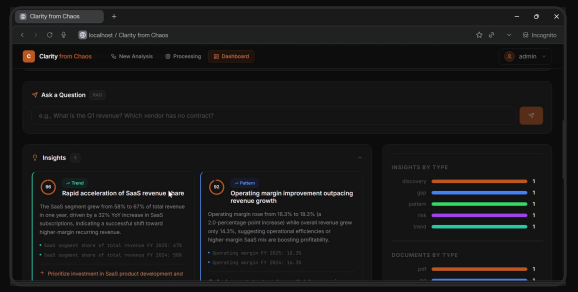
End-to-End User Journey



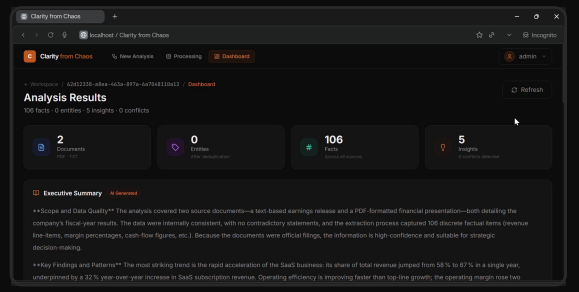
Upload multi-format files



Real-time agent stream + LLM reasoning



AI-generated insights with confidence scores



Executive summary + RAG Q&A

Upload → 8-Stage Pipeline → Dashboard → Ask Questions in Plain English

FUTURE SCALABILITY

Roadmap & Expansion Opportunities

Designed from day one to scale across data, models, and teams.

01 — Near Term

Scale the Data

- Incremental ingestion — add docs to existing jobs
- Streaming ingestion for live data feeds (Kafka)
- Cloud storage backends: S3, Azure Blob, GCS
- Multi-tenancy with per-user job isolation

02 — Mid Term

Scale the Intelligence

- Swap embeddings for sentence-transformers (GPU)
- Fine-tuned models for domain-specific entity extraction
- Temporal conflict tracking (v1 vs v2 of same doc)
- Automated report scheduling & email delivery

03 — Long Term

Scale the Platform

- Kubernetes deployment with horizontal autoscaling
- Celery + Redis task queue for async pipelines
- Plugin architecture — add custom agent nodes
- Enterprise SSO, audit logs, data lineage tracking

"From chaos to clarity — built to grow with your data."